Classification and Clustering of Big Data: Online Retail Customers

Assignment No. 2

1. Execution environment and scope

All code in this report was executed locally on a Windows 11 machine with Python 3.12, pandas, scikit-learn and matplotlib. The dataset is the UCI Online Retail dataset (541,909 raw rows). The Kaggle datasets listed in the brief were not used. Streaming clustering is an incremental mini-batch k-means approximation, and subspace clustering uses PCA plus an exhaustive two-feature search. Neither is a full CluStream, StreamKM++, CLIQUE or PROCLUS implementation.

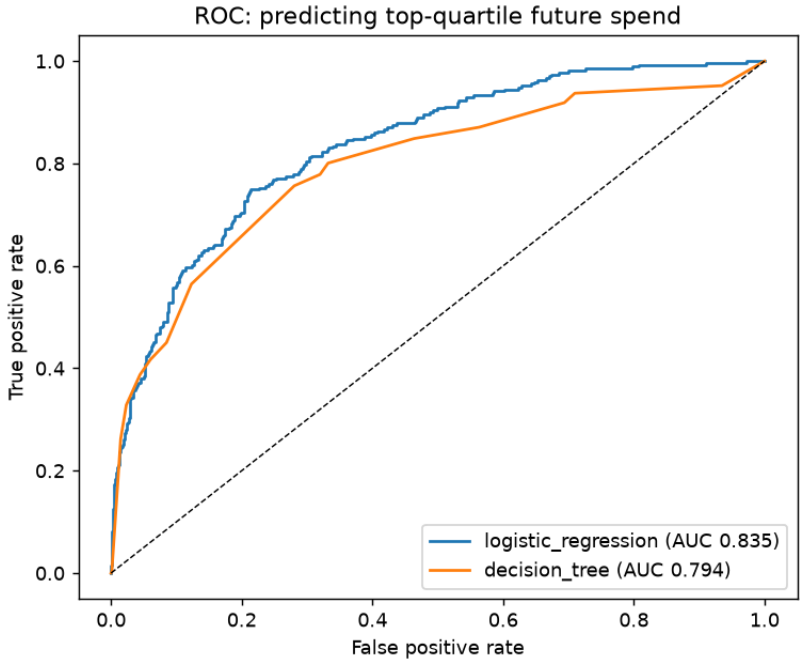
2. Data and labels

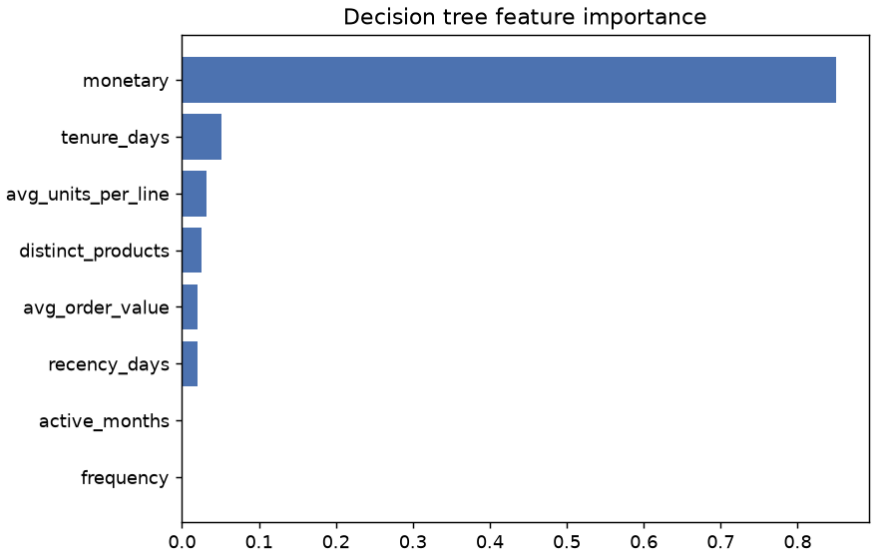
After cleaning: 391,150 rows, 4,334 customers, 18,402 invoices. Features come from transactions before 1 Oct 2011. The label is *high value*: a customer's spend after 1 Oct 2011 is in the top quartile (threshold 516.24). Using a time split avoids leakage between inputs and target. 3,613 customers are in the dataset, and 904 are labelled high value.

3. Supervised classification

Model	Accuracy	Precision	Recall	F1	ROC AUC
logistic_regression	0.790	0.570	0.642	0.604	0.835
decision_tree	0.705	0.448	0.779	0.569	0.794

Train 2,529, test 1,084, positive rate 25.02%. Both models use balanced class weights. Logistic regression is the stronger model on AUC and F1. The decision tree (depth 5) has higher recall but lower precision, and its feature importances show which behaviours matter most.




Decision tree feature importance
monetary
tenure_days
avg_units_per_line
distinct_products
avg_order_value
recency_days
active_months
frequency
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

4. Unsupervised clustering

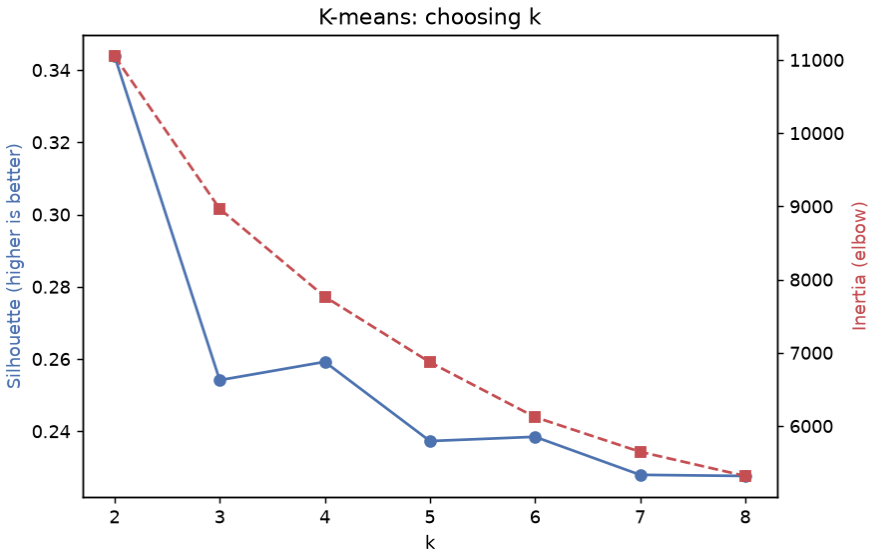
Features: recency, frequency, monetary, average order value and distinct products, log-transformed and standardized. K-means chose k by silhouette score. DBSCAN used eps set at the 90th percentile of 5-nearest-neighbour distances.

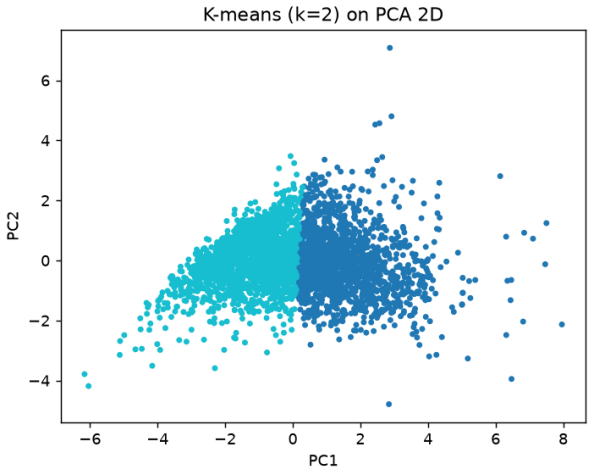
k	Inertia	Silhouette	Davies-Bouldin
2	11050	0.344	1.131
3	8967	0.254	1.288
4	7766	0.259	1.281
5	6871	0.237	1.245
6	6122	0.238	1.212
7	5648	0.228	1.235
8	5318	0.228	1.219

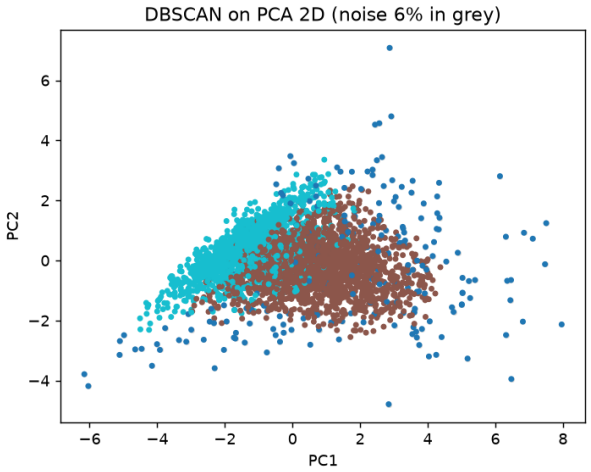
The best silhouette is at k=2. DBSCAN found 2 clusters with 6.0% noise (eps 0.688, min_samples 5). The clusters form a continuum rather than separate groups, so the two-cluster solution is a broad split, not distinct segments.

Cluster profiles (medians):

Cluster	Customers	Recency (days)	Frequency	Monetary	Avg order	Products	Future spend
0	1563	22	4	1610	381	70	437
1	2050	122	1	306	211	17	0



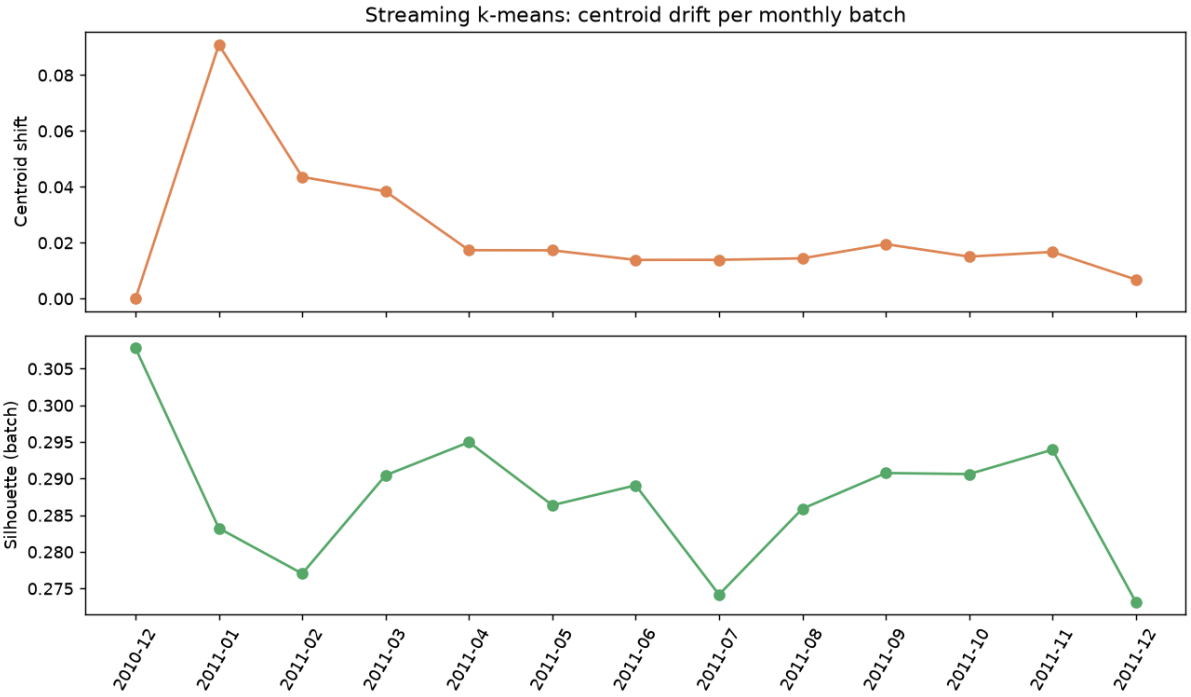

K-means (k=2) on PCA 2D
PC2
PC1
6
4
2
0
−2
−4
−6
−4
−2
0
2
4
6
8


DBSCAN on PCA 2D (noise 6% in grey)
PC2
PC1
6
4
2
0
−2
−4
−6
−4
−2
0
2
4
6
8

5. Streaming clustering

Invoices are processed month by month in time order. MiniBatchKMeans (k=4) is updated with partial_fit on each monthly batch, and centroid drift and silhouette are recorded. The scaler is fixed on the first batch. This is incremental clustering, not CluStream or StreamKM++.

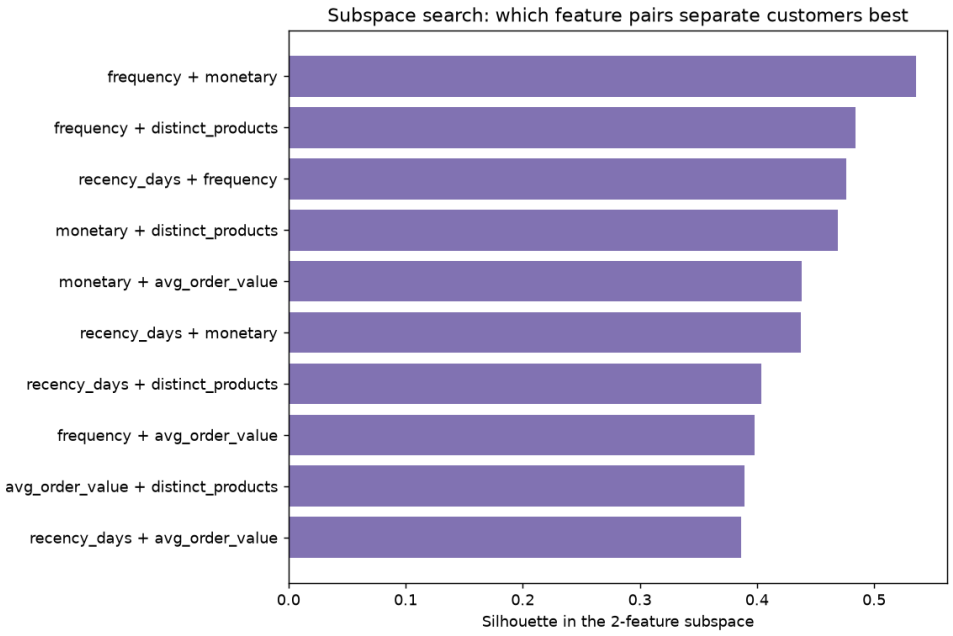
Month	Invoices	Centroid shift	Silhouette
2010-12	1,394	0.0000	0.308
2011-01	983	0.0909	0.283
2011-02	992	0.0436	0.277
2011-03	1,312	0.0384	0.290
2011-04	1,139	0.0174	0.295
2011-05	1,544	0.0173	0.286
2011-06	1,390	0.0139	0.289
2011-07	1,321	0.0139	0.274
2011-08	1,267	0.0144	0.286
2011-09	1,739	0.0195	0.291
2011-10	1,903	0.0150	0.291
2011-11	2,642	0.0167	0.294
2011-12	776	0.0068	0.273



6. Feature extraction and subspace clustering

PCA keeps 90% of the variance with 3 components. Silhouette at k=2: full feature space 0.344, PCA subspace 0.373. The PCA subspace is slightly better, meaning a few combined directions carry the separation. The search over all two-feature subspaces ranks which pairs separate customers best.

Subspace (2 features)	Silhouette
frequency + monetary	0.536
frequency + distinct_products	0.484
recency_days + frequency	0.476
monetary + distinct_products	0.469
monetary + avg_order_value	0.438
recency_days + monetary	0.437
recency_days + distinct_products	0.404
frequency + avg_order_value	0.398
avg_order_value + distinct_products	0.390
recency_days + avg_order_value	0.386



7. Collaborative filtering

Item-item cosine similarity on the top 500 products, using purchases before 1 Oct 2011. For 1,000 customers with purchases before and after the cutoff, the top 5 unseen products are recommended and compared with the products each customer bought later. The popularity baseline recommends the most-bought products the customer has not yet bought.

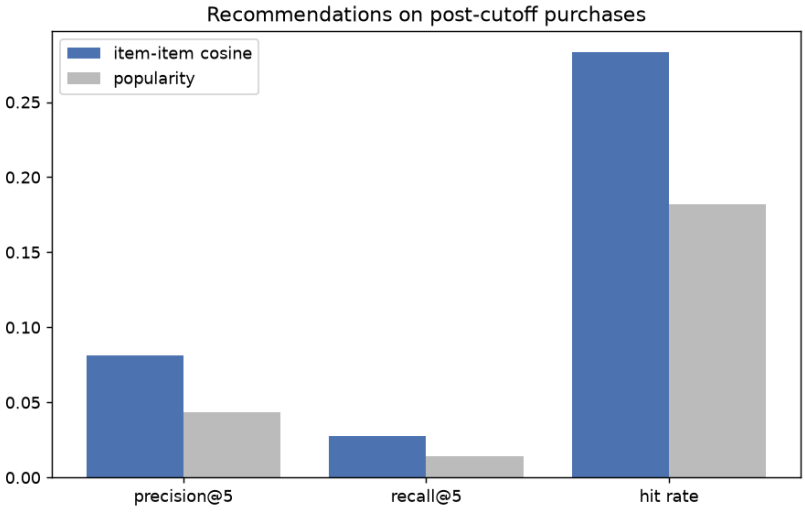
Method	Precision@5	Recall@5	Hit rate
Item-item cosine	0.081	0.027	0.283
Popularity baseline	0.044	0.014	0.182

Example recommendations:

Customer 12347: bought earlier: REGENCY CAKESTAND 3 TIER, ROSES REGENCY TEACUP AND SAUCER, ALARM CLOCK BAKELIKE RED. Recommended: PACK OF 72 RETROSPOT CAKE CASES, GREEN REGENCY TEACUP AND SAUCER, SET OF 3 REGENCY CAKE TINS, LUNCH BAG RED RETROSPOT, LUNCH BAG DOILEY PATTERN.

Customer 12352: bought earlier: REGENCY CAKESTAND 3 TIER, SET OF 3 CAKE TINS PANTRY DESIGN, SPOTTY BUNTING. Recommended: RECIPE BOX PANTRY YELLOW DESIGN, JAM MAKING SET WITH JARS, PACK OF 72 RETROSPOT CAKE CASES, JAM MAKING SET PRINTED, WHITE HANGING HEART T-LIGHT HOLDER.

Customer 12356: bought earlier: REGENCY CAKESTAND 3 TIER, PACK OF 72 RETROSPOT CAKE CASES, ROSES REGENCY TEACUP AND SAUCER. Recommended: PACK OF 60 PINK PAISLEY CAKE CASES, PACK OF 60 DINOSAUR CAKE CASES, SET/10 RED POLKADOT PARTY CANDLES, PACK OF 60 SPACEBOY CAKE CASES, GINGERBREAD MAN COOKIE CUTTER.



8. Conclusions and limitations

- Logistic regression separates high-value customers reasonably well (AUC shown in section 3).

- Clusters are a continuum on these features. K-means at k=2 is a broad split, not a set of distinct segments.

- Streaming results are from an incremental approximation. Centroid drift is largest in the first month, then settles.

- Subspace clustering uses PCA and a two-feature search, not a dedicated subspace algorithm.

- Collaborative filtering beats the popularity baseline on hit rate and recall. Absolute accuracy is low, as expected for implicit purchase data.

- The Kaggle datasets were not used. Only the UCI Online Retail data was analysed.